Streamlining IT Procurement: Automating Standard Laptop Orders with WI

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Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with WI
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1. Abstract

This project focuses on streamlining the IT procurement process for standard laptop orders in an organisation. Currently, requesting a new laptop involves many manual steps such as filling paper forms, waiting for manager approval, checking budget availability, selecting vendors, and tracking inventory. These steps cause delays and errors. Our proposed solution uses Workflow Intelligence (WI) to automate the complete laptop order process. Employees can raise a procurement request through a simple web portal. The system then automatically routes the request for manager approval, performs a budget check, selects the best vendor based on predefined rules, updates inventory, and places the order. The goal is to reduce processing time, improve accuracy, and give better visibility of every order status.

2. Introduction

Every organisation needs to provide laptops to new joiners and also replace old machines for existing employees. The traditional way of handling laptop procurement is mostly manual. An employee or HR raises a request, the manager approves it on email or paper, the finance team checks the budget, the purchase team contacts vendors, and finally the inventory is updated after the laptop arrives. This process can take several days or even weeks.

With the growth of digital tools, it is possible to automate most of these steps using Workflow Intelligence (WI). WI means using software that understands the business rules and moves the request from one stage to the next without human intervention where possible. This project designs and describes such an automated system for standard laptop orders. The system covers the full cycle from request submission to order placement and inventory update.

3. Problem Statement

The existing laptop procurement process faces several practical problems:

- Manual form filling and email-based approvals lead to lost requests and long waiting times.
- Managers and finance teams have no single place to see pending approvals or budget status.
- Budget checks are done late in the process, causing rejections after a lot of effort is already spent.
- Vendor selection is done manually and is often based on personal preference rather than price or delivery time.
- Inventory records are updated only after the laptop arrives, so the team does not know real-time stock levels.
- There is no clear tracking for the employee who raised the request; they keep asking for status updates.

Because of these issues, the average time to deliver a standard laptop is high, employee satisfaction is low, and the IT and purchase teams spend too much time on routine work instead of higher-value tasks.

4. Objectives

The main objectives of this project are:

- To design an automated workflow for standard laptop procurement requests using Workflow Intelligence (WI).
- To provide a simple web-based interface where employees can raise laptop requests and track status.
- To implement automatic manager approval routing based on department and cost centre rules.
- To perform real-time budget check before the request moves to vendor selection.
- To select vendors automatically using predefined criteria such as price, delivery time and preferred vendor list.
- To keep inventory updated and reduce stock-out situations for standard laptop models.
- To reduce the overall cycle time of laptop orders from several days to a few hours where possible.

5. Existing System

In the current system, the laptop request process is largely offline or email-driven. An employee or the HR team sends a request by email or fills a shared spreadsheet. The manager receives the mail and replies with approval or rejection. After approval, the request is forwarded to the finance team for budget confirmation. Only after budget clearance does the purchase team start contacting vendors one by one. Inventory is maintained in a separate Excel file or in a basic inventory tool that is not linked to the request process.

There is no single system that connects request, approval, budget, vendor and inventory. As a result, the process depends heavily on individual follow-ups. Status is hard to track and reports are prepared manually at the end of the month. This existing approach works for a small number of requests but becomes inefficient when the organisation grows and the number of laptop requests increases.

6. Proposed System

The proposed system introduces a complete automated workflow for standard laptop orders. Employees log into a web portal and submit a procurement request by selecting the laptop model, quantity and reason (new joiner / replacement). The system immediately checks the current inventory. If stock is available, it reserves the laptop and notifies the employee and the IT asset team. If stock is not available, the request enters the procurement workflow.

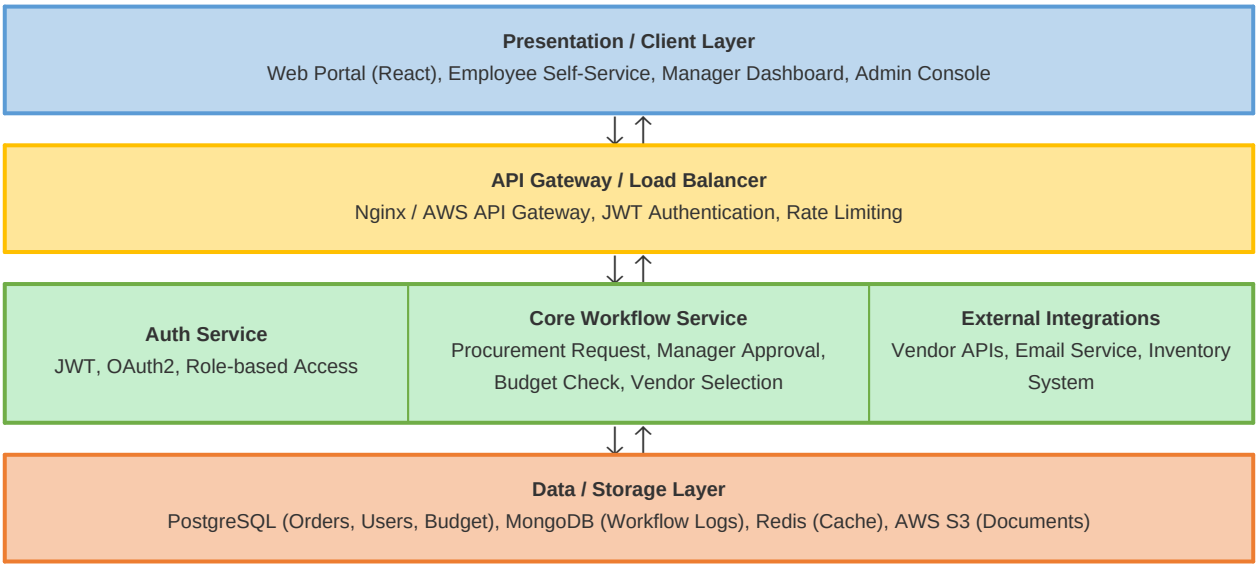
The workflow engine automatically routes the request to the correct manager based on the employee's department and reporting structure. After manager approval, a budget check is performed against the department cost centre. If the budget is sufficient, the system selects a vendor from the approved vendor list using rules such as lowest price for the standard model or shortest delivery time. A purchase order is generated and sent to the vendor. At the same time, the employee, manager and purchase team receive status notifications. When the laptop is delivered, inventory is updated and the asset is assigned to the employee.

All steps are logged, so any authorised user can see the current stage of any request. This removes the need for repeated follow-up emails and gives management clear visibility of procurement performance.

7. System Architecture & Workflow

7.1 Solution Architecture Diagram

The architecture follows a layered design with clear separation between the user interface, API layer, business services and data storage. The diagram below shows the main components used in the solution.



7.2 Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for employees to raise laptop requests, for managers to approve or reject, and for admins to configure rules and view reports.	React.js, HTML5, CSS3, REST APIs
API Gateway	Acts as the single entry point for all client requests. Handles authentication, load balancing, request routing and basic security checks.	Nginx / AWS API Gateway, JWT
Auth Service	Manages user login, issues secure tokens, and enforces role-based access control so that only authorised users can perform specific actions.	JWT, OAuth2, Node.js / Spring Security
Core Workflow Service	Contains the main business logic for the automated workflow: creating procurement request, routing for manager approval, performing budget check, selecting vendor and generating purchase order.	Node.js / Java Spring Boot, Workflow Engine rules
External Integrations	Connects to vendor systems for price and availability, sends email notifications, and updates the inventory management system after order completion.	REST APIs, SMTP Email Service, Vendor APIs
Data / Storage Layer	Stores all persistent data including user profiles, requests, approval history, budget data, vendor master, inventory levels and workflow audit logs.	PostgreSQL, MongoDB, Redis, AWS S3

7.3 Step-by-Step Laptop Order Automation Workflow

The automated workflow for a standard laptop order works as follows:

Step 1 – Request Submission: Employee logs into the web portal, selects the standard laptop model, enters quantity and reason (new joiner or replacement), and submits the procurement request.

Step 2 – Inventory Check: The system immediately queries the inventory database. If the required laptop is available in stock, it is reserved and the request is marked as fulfilled from inventory. The process ends here for that request.

Step 3 – Manager Approval Routing: If stock is not available, the workflow engine identifies the correct reporting manager using the employee's department and organisation hierarchy. An approval task is created and the manager receives a notification.

Step 4 – Manager Decision: The manager reviews the request on the dashboard and either approves or rejects it. In case of rejection, the employee is notified with the reason and the workflow stops.

Step 5 – Budget Check: After approval, the system checks the available budget for the employee's cost centre. If the budget is insufficient, the request is sent back or escalated to finance. If the budget is enough, the amount is reserved.

Step 6 – Vendor Selection: The system applies vendor selection rules (for example: preferred vendor list, lowest quoted price for the standard model, or shortest promised delivery time) and selects the most suitable vendor automatically.

Step 7 – Purchase Order Generation: A purchase order is generated with the selected vendor details, laptop model, quantity and expected delivery date. The PO is sent to the vendor through the integration layer.

Step 8 – Notification & Tracking: Status updates are sent by email and shown on the portal to the employee, manager and purchase team. The request status remains visible until the laptop is delivered.

Step 9 – Delivery & Inventory Update: When the vendor delivers the laptop, the IT asset team marks the order as received. Inventory is updated, the asset is assigned to the employee, and the workflow is closed.

8. Benefits

The automated system provides the following benefits:

- Faster processing – standard laptop orders can move from request to purchase order within hours instead of days.
- Reduced manual effort – managers, finance and purchase teams spend less time on routine follow-ups.
- Better budget control – budget is checked early and reserved, reducing the chance of overspending.
- Consistent vendor selection – rules ensure the same fair criteria are applied every time.
- Real-time inventory visibility – the system always knows current stock levels for standard models.
- Full audit trail – every approval, rejection and status change is recorded for compliance and reporting.
- Improved employee experience – requestors can see the status of their laptop request at any time.

9. Conclusion

This project shows that the standard laptop procurement process can be significantly improved by introducing Workflow Intelligence (WI). By connecting the procurement request, manager approval, budget check, vendor selection and inventory update into one automated flow, organisations can reduce delays, cut manual work and gain better control over IT asset spending. The layered architecture keeps the system modular so that individual services can be updated or replaced without affecting the whole solution. Overall, the proposed system offers a practical and scalable way to modernise a common but often inefficient business process.

10. Future Scope

The current design can be extended in several useful directions:

- Support for non-standard and high-value laptop models with multi-level approval chains.
- Integration with HR systems so that new joiner laptop requests are created automatically on joining date.
- Mobile application for managers to approve requests on the go.
- Predictive analytics to forecast laptop demand based on hiring plans and replacement cycles.
- Expansion of the same workflow pattern to other IT assets such as monitors, headsets and software licences.

• Addition of a chatbot interface so employees can raise and track requests using natural language.

11. References

1. Official documentation of React.js and modern web application design patterns.
2. AWS API Gateway and serverless architecture best practices for enterprise applications.
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4. JWT and OAuth2 standards for secure authentication and authorisation.
5. PostgreSQL and MongoDB documentation for relational and document data storage.
6. Industry articles on digital transformation of procurement and IT asset management.
7. SmartBridge SkillWallet project guidelines and architecture template references.